

# 0602221- AI Programming

## HW 03

### Main Topics

- User-validated input
- `if` and `if-else` statements
- Loop statements

### Program Specification

Write a Python program that allows a single player (the user) to play a simple four-dice game of chance against **The Odds**.

### Game Description

- There is a single player with four four-sided dice.
  - The sides of each die are labeled with the numbers from 1 to 4.
  - The value of each die is a randomly generated integer from 1 to 4.
  - A game is made up of rounds. A single round is played as follows:
1. The player rolls four dice. In Python, each die can be generated using:

```
import random
```

```
d1 = random.randint(1, 4)
d2 = random.randint(1, 4)
d3 = random.randint(1, 4)
d4 = random.randint(1, 4)
```

2. Display the four dice in a reasonable format.

3. Determine whether the player has won the round.

The player loses the round unless one of the following occurs:

- **Quad:** all four dice have the same value.
- **Triple:** exactly three of the four dice have the same value.
- **Two Pair:** two dice have one value and the other two dice have a different matching value.
- **High Pair:** exactly two dice show the value 4.

4. Report the result of the round as one of the following:

- Quad
- Triple
- Two Pair
- High Pair
- Loss

5. Ask the player if they wish to play another round.

6. When the player chooses not to continue, display a final report showing:

- total number of rounds played
- number of rounds won
- number of rounds lost

## Requirements

- Use Python.
- Use `if / elif / else` statements to determine the round result.
- Use a loop to allow multiple rounds.
- Validate the user's choice when asked whether to play again.
- Keep track of total rounds, rounds won, and rounds lost.

## Sample Run(s)

```
Welcome to Computer Dice
```

```
-----
```

```
You will first roll your dice
Next the outcome of your roll will
be determined:
```

any Quad and you Win  
any Triple and you Win  
any Two Pair and you Win  
any High Pair (4's) and you Win  
anything else and you Lose

-----

Player

-----

2 2 4 4  
Two Pair, you win!  
Do you wish to play again [y, n] : y

Player

-----

1 1 4 3  
Sorry, you lose!  
Do you wish to play again [y, n] : y

Player

-----

4 4 3 1  
High Pair, you win!  
Do you wish to play again [y, n] : y

Player

-----

3 3 3 3  
Quad, you win!  
Do you wish to play again [y, n] : n

Computer Dice Results

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You played 4 rounds  
Rounds won : 3  
Rounds lost : 1

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## Grading

| Performance Indicator         | [1] | [2] | [3] |
|-------------------------------|-----|-----|-----|
| Readability and documentation | 0   | 1   | 1   |
| Use of if statements          | 1   | 2   | 2   |
| Use of while loops            | 1   | 2   | 2   |
| Functional requirements       | 2   | 3   | 4   |
| Efficiency                    | 0   | 1   | 1   |

## Submission

Submit your Python file according to your instructor's submission instructions.

A suggested file name is:

hw3\_ID#.py